RAMAIAH UNIVERSITY OF APPLIED SCIENCES

Instructions to students:

1. Maximum marks: **10**
2. The assignment has to be neatly word processed as per the prescribed format.
3. The maximum number of pages should be restricted to **8**
4. The printed assignment must be submitted to the course leader.
5. **Submission Date: 16/11/2026**

Assignment

Question 1 **(10 Marks)**

Develop a multithreaded banking/ATM application in which multiple customers simultaneously access the bank's services through ATMs and online banking platforms.

The application should support concurrent execution of various banking transactions, such as withdrawal, deposit, balance enquiry, and fund transfer. Since multiple threads may access and modify the same bank account concurrently, the system should demonstrate potential concurrency-related problems, including race conditions, inconsistent account balances, and deadlocks.

The application should further demonstrate the use of appropriate synchronization mechanisms to prevent these problems and ensure that all banking transactions are executed safely, consistently, and correctly under concurrent execution.

The document should contain the following:

1.1: Design of banking/ATM application

1.2: Implementation of the application

1.3: Testing and validation of the application

CO – Course Outcome

Course Outcomes	
1	Explain the core and advanced functionalities of an operating system..
2	Apply operating system concepts to meet various computing requirements.
3	Analyze operating system services for different computing scenarios.
4	Evaluate various operating system resource management techniques.